

Lesson 01 worksheet

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1. Solve $\sin(\varphi + \pi/3) = -1/2$ where φ in radians and $-\pi \leq \varphi \leq \pi$. Give answers in terms of π . [4 marks]

2. Solve equation $3\operatorname{cosec}^2\theta - 5 = 5\cot\theta$ for $0^\circ \leq \theta \leq 180^\circ$ [4 marks]

3. Solve the equation $\cos^2(x)\tan(x) - (1/2)\tan(x) = 0$ for $-\pi < x < \pi$ [5 marks]

4. Hence solve the equation $6\sin^3(\theta) + \sin^2(\theta) - 12\sin(\theta) + 5 = 0$ for $0^\circ \leq \theta \leq 90^\circ$ [2 marks]

5. Find the value of a, of b and of c in $y = a \sin(x/b) + c$ [3 marks]

6. Show that $\sin^3(x)(\operatorname{cosec}(x)/\cot(x))$ can be written as $\sin^2(x)\tan(x)$ [3 marks]

Answer key

1. $\phi = -\pi/2, \phi = 5\pi/6$
2. $\theta = 108.4^\circ, \theta = 26.6^\circ$
3. $x = 0, \pm\pi/4, \pm3\pi/4$
4. $\theta = 30^\circ, \theta = 90^\circ$
5. $a=4, b=3, c=-2$
6. Correct simplification using trig identities to reach $\sin^2(x)\tan(x)$